



Republic of Rwanda
Ministry of Education



GENBN401

BASICS OF NETWORKING

Perform Basics of Networking

Competence

RQF Level: 4

Learning Hours



70

Credits: 7

Sector: ICT and Multimedia

Trade: Software Development

Module Type: General

Curriculum: ICTSWD4002- TVET Certificate 4 in Software Development

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1200

Issue Date: August 2023

Purpose statement	This module Describes the skills, Knowledge and attitude required to Perform basics of Networking. This module intended to prepare students Pursuing TVET in Level 4 Software Development. At the end of this module, the students will be able to Establish network media connectivity, Perform Basic Network Configuration, Maintain Network system						
Learning assumed to be in place	N/A						
Delivery modality	Training delivery			100%	Assessment		Total 100%
	Theoretical content			30%	Formative assessment	20%	100%
	Practical work:			70%			
	Group project and presentation	30%	80%				
	Individual project /Work	40%					
				Summative Assessment			

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1.Establish network media connectivity	1.1. Tools, materials, and equipment are correctly identified based on network requirements.
	1.2. Network cables are perfectly terminated based on cabling types
	1.3. Network media are properly connected based on Network topology
	2.1. IP addresses are correctly classified based on their types and versions

2. Perform Basic Network Configuration	2.2. IP addresses subnet masks are appropriately calculated based on the network topology
	2.3. IP addresses are appropriately assigned according to the network topology.
	2.4. Network devices are correctly configured based on manufactures' guide
	2.5. Interconnectivity is correctly tested according to the configured network Functionalities
3. Maintain Network system	3.1 Preventive maintenance is properly checked as per manufacturer's guidelines
	3.2. Corrective Maintenance measures are applied based on problems identified.
	3.3. Maintenance report is properly elaborated based on the work done

Intended Knowledge, Skills, and Attitudes

Knowledge	Skills	Attitudes
• Describe Network Devices • Describe Network Services • Identify Network media • Describe IP Addressing • Identify Principles of subnetting • Describe Network Security Basics • Describe Network Troubleshooting	• Perform Cabling and Physical Setup • Assign and configure IP addresses on devices • Configure Network devices • Perform Subnetting • Use Basic Command Line • Perform Basic Network Troubleshooting • Perform Network Documentation • Communication Skills • Problem-Solving Skills	• Curiosity • Patience • Persistence • Critical Thinking • Attention to Detail • Adaptability • Collaborative Spirit • Ethical Behaviour • Resilience • Continuous Learning • Empathy • Security Consciousness • Resourcefulness • Ownership and Accountability

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Establish network media connectivity 2. Perform Basic Network Configuration 3. Maintain Network system
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Learning outcome 1: Establish network media connectivity	Learning hours:20
Indicative content	
• Identification of Network requirements. ✓ Description of network concepts and technologies  Basic definitions  Network classifications  Network benefits  Advantages and Disadvantages of network  Application of network  Network technologies  Network topology types  Network components ✓ Materials  Network Cables (twisted, coaxial, management and Fiber optic)  Trunk (Flexible, plastic, timber, and stainless steel)  Connectors  Cable Ties  Cable clips  Cable Sockets	

-  Wall plugs

- ✓ Tools

-  Cutting Tools

-  Stripping tools

-  Drilling Tools

-  Fixing Tool

-  Patching Tools

-  Crimping tools

-  Testing tool

- ✓ Equipment

-  Computer

-  Inverter

-  UPS

-  Inverter

-  Switch

-  Glue gun

-  Rack

-  Brackets

-  Patch panel

-  Repeater

-  Regenerator

- **Termination of Network cables**

- ✓ Network cables installation types.

-  Open-Wire

-  Aerial

-  Above-Grounds Conduits

-  Underground

-  Underwater

-  Built in

✚ Semi built in ✓ Network cables Trunking materials ✚ Plastic ✚ Wood ✚ Stainless ✓ Cable termination ✚ Twisted pair cabling ✚ Fiber-optic cabling ✚ Coaxial cabling ✚ Shielded twisted pair • Connection of Network Media ✓ Labelling ✓ Patching and Tagging ✓ Provide us build design	
Resources required for the learning outcome	
Equipment	▪ Computer ▪ Inverter ▪ Switch ▪ Battery ▪ Firewalls ▪ Rack ▪ UPS
Materials	▪ Internet bundles ▪ Network cables ▪ Electricity
Tools	▪ Networking toolkit ▪ Simulation tools

Facilitation techniques	▪ Demonstration and simulation ▪ Individual and group work ▪ Practical exercise ▪ Individualized ▪ Group discussion
Formative assessment methods / (CAT)	▪ Written assessment ▪ Performance assessment ▪ Oral presentation

Learning outcome 2: Perform Basic Network Configuration	Learning hours: 35
Indicative content	
• Classification of IP Addresses ✓ Types of IP Addresses ✓ IP address versions ✓ Identification of IP address classes • Calculation of IP addresses subnet masks ✓ Introduction to subnet masks ✚ Definition ✚ Benefits of subnetting ✓ Binary system ✓ Types of Subnetting ✓ Logical and bitwise Operators • Assigning IP Address ✓ Static ✓ Dynamic ✓ Automatic • Configuration of Basics Network Device. ✓ Device Configuration Modes	

✓ Host name ✓ Banner message ✓ Reload Device ✓ Configure port ✓ Configure Device passwords ✓ Save configuration • Testing network Interconnection ✓ Physical Testing ✓ Unit Testing ✓ Integration Testing	
Resources required for the learning outcome	
Equipment	Computer Inverter Battery UPS
Materials	Network cables Connectors Flexible PIPE Cables Cables Ties Cables clips
Tools	Networking toolkit Drilling tools Fixing tool Pliers
Facilitation techniques	Demonstration and simulation Individual and group work Practical exercise Individualized

	Trainer guided Group discussion
Formative assessment methods /(CAT)	Written assessment Oral presentation Performance assessment Product based assessment

Learning outcome 3: Maintain Network system	Learning hours: 15
Indicative content	
• Perform preventive maintenance. ✓ Hardware preventive maintenance + Schedule regular cleaning + Setting of preventive measures + Check physical Equipment condition. + Check environment condition. ✓ Software preventive maintenance + Regular change of network device credentials + Network monitoring software Licencing /Application + Updating and Upgrading network monitoring software and device firmware • Perform corrective maintenance. ✓ Hardware corrective maintenance + Identification of common problem and their causes + Repair/Replace damaged equipment. ✓ Software corrective maintenance + Troubleshoot network configuration. + Check network status + Update network configuration • Troubleshooting network	

- ✓ Introduction to troubleshoot
- ✓ Troubleshoot process
 - ✚ Collecting Network System information
 - ✚ Analysing current Network Status
 - ✚ Identification of common problem
 - ✚ Implementation of solution

• **Elaboration of maintenance report**

- ✓ Ways of reporting
 - ✚ Oral
 - ✚ Written
 - ✚ Video documentation
- ✓ Report elements
 - ✚ Used Tools, materials, and Equipment.
 - ✚ Status after maintenance
 - ✚ Update us built design.
 - ✚ Recommendation

Resources required for the indicative content

Equipment	▪ Computer ▪ Router ▪ battery ▪ Visual Equipment
Materials	▪ Cables and accessories ▪ Electricity ▪ Internet bundles ▪ Power Extension
Tools	▪ Network Toolkits

Facilitation techniques	▪ Demonstration and simulation ▪ Individual and group work ▪ Practical exercise ▪ Individualized ▪ Trainer guided ▪ Group discussion
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment ▪ Product based assessment

References

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2. Dionys, D. (2008-2013). *How to Make a Network Cable*. Unpublished. Produced for the VVOB Program.
3. John Wiley & Sons, I. (2017). *Microsoft Technology associate exam 98-366*. Microsoft Corporation.
4. *Local area networks — how computers talk to each other*. (1984). Institution of Electrical Engineers.
5. M., C. (1984). *Local area networks*. *British Library*. Library and Information Research Report 19.
6. Mitchell, B. (2010). *Introduction to Client Server Networks*. from About.com: <http://compnetworking.about.com/od/basicnetworkingfaqs/a/client-server.htm>.